

$$e^{-x} - x = 0; \quad x - \tan x = 0; \quad \dots$$

• v splošnem o rešitvah ne moremo reči ničesar

Metode reševanja

bisekcija

iteracija

$$f(x) = 0 \rightarrow x = g(x); \quad x_0 \text{ zač. pribl.} \rightarrow x_{r+1} = g(x_r)$$

$$g(x) = x - f(x); \quad g(x) = x - cf(x); \quad c \neq 0; \quad g(x) = x - h(x)f(x); \quad h(x) \neq 0$$

• Če $g(x)$ skrčitev na $[\alpha - \delta, \alpha + \delta]$; zaporedje $x_{r+1} = g(x_r)$ konvergira k α (α je negibna točka ($g(\alpha) = \alpha$)). Velja $|x_r - \alpha| \leq m^r |x_0 - \alpha|$;

• $g(\alpha) = \alpha$; g zv. odv. v α . Če $|g'(\alpha)| < 1$,

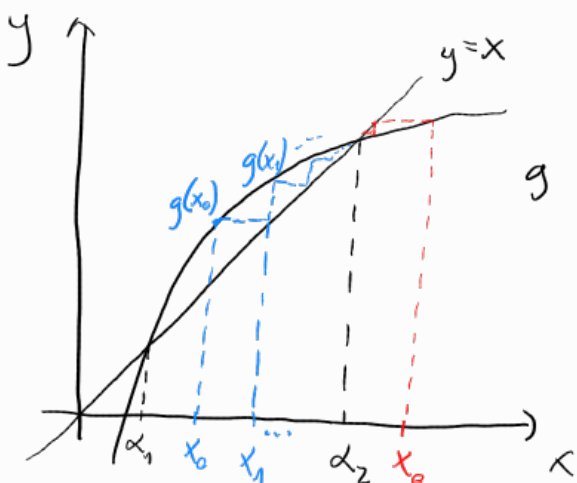
$\exists$ I okolica α , da $x_{r+1} = g(x_r)$ konv. k α .

($m = |g'(\alpha)|$, uporabimo prejšnjo trditev)

$$|x_{r+1} - \alpha| \leq \frac{m}{1-m} |x_r - x_{r-1}|.$$

$$(0 < m < 1)$$

↑
iz pogoja za skrčitev



$g'(\alpha_1) > 1$ odbojna
 $g'(\alpha_2) < 1$ privlačna

• Naj x_r konv. k α . Red konvergence je p , če $\exists C_1, C_2 > 0$: za dovolj
pozne člene zap. velja $C_1 |x_r - \alpha|^p \leq |x_{r+1} - \alpha| \leq C_2 |x_r - \alpha|^p$.

npr. če $|x_r - \alpha| \approx 10^{-3}$, bo napaka $|x_{r+1} - \alpha| \approx 10^{-3p}$

- g p -krat zv. odv. vok. α ; $g^{(k)}(\alpha) = 0$, $k=1, \dots, p-1$ in $g^{(p)}(\alpha) \neq 0$. Potem ima g p -ti red konv. vok. α .

Tangentna metoda

$$x_{r+1} = x_r - \frac{f(x_r)}{f'(x_r)}$$

- primer navadne iteracije,

$$g(x) = x - \frac{f(x)}{f'(x)}$$

- konvergenca: vsaj reda 2

1) α je enostavna ničla ($f'(\alpha) \neq 0$):

$$i) g(\alpha) = \alpha - \frac{f(\alpha)}{f'(\alpha)} = \alpha$$

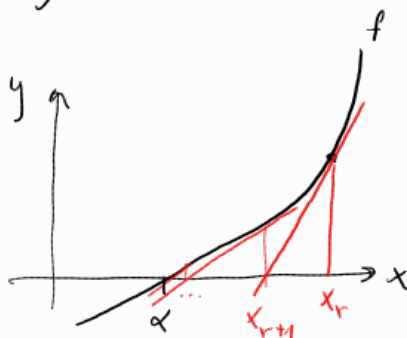
$$ii) g'(\alpha) = 1 - \frac{f'(\alpha)^2 - f(\alpha)f''(\alpha)}{f'(\alpha)^2} = \frac{f(\alpha)f''(\alpha)}{f'(\alpha)^2}$$

$$g'(\alpha) = 0 \quad \checkmark$$

$$g''(\alpha) = \frac{(f'f'' + ff''')f'^2 - 2ff''f'''}{f'^4}$$

$$g''(\alpha) = \frac{f''(\alpha)}{f'(\alpha)} \Rightarrow \text{če } f''(\alpha) = 0, \text{ je red konv. vsaj } 3 \\ \neq 0, \text{ je red konv. } 2$$

geometrično:



analitično:

Taylor za $f(x_r + \Delta x_r) = 0$:

$$0 \doteq f(x_r) + f'(x_r)\Delta x_r + \dots$$

$$x_{r+1} - x_r = \Delta x_r = -\frac{f(x_r)}{f'(x_r)} \rightarrow x_{r+1} = x_r - \frac{f}{f'} \quad \checkmark$$

2) α je m -kratna ničla, $m > 1$

$$f(x) = (x-\alpha)^m h(x); \quad h(\alpha) \neq 0$$

$$g(x) = x - \frac{f(x)}{f'(x)} = x - \frac{(x-\alpha)^m h(x)}{m(x-\alpha)^{m-1} h(x) + (x-\alpha)^m h'(x)} = x - \frac{(x-\alpha) h(x)}{m h(x) + (x-\alpha) h'(x)}$$

$$g'(\alpha) = 1 - \frac{m h(\alpha)^2}{m^2 h(\alpha)^2} = 1 - \frac{1}{m}$$

Sekantna metoda

Problem tangentne metode: odvod zoprn za računati

$$\rightarrow f'(x_r) \doteq \frac{f(x_r) - f(x_{r-1})}{x_r - x_{r-1}}$$

$$x_{r+1} = x_r - \frac{f(x_r)(x_r - x_{r-1})}{f(x_r) - f(x_{r-1})} ; \rho = \frac{\sqrt{5} + 1}{2} \doteq 1,62$$